AMANUEL AYALEW

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SUMMARY

Software Engineer with experience in backend systems and growing expertise in Machine Learning, with a focus on research-driven development. Experience in building and improving scalable applications, Skilled in experimentation, performance analysis, and exploring new approaches to model design. Passionate about applying machine learning systems to solve real-world, data-driven problems.

EDUCATION

Bachelor of Science in Software Engineering 2021 – 2026
Addis Ababa University (AAiT)

PROFESSIONAL EXPERIENCE

Icog Labs July 2025 – May 2026
ML Engineer Intern

- Conducted research on predictive coding–based transformer architectures as alternatives to backpropagation
- Worked on a tag recommendation model for the Mindplex platform, improving recommendation accuracy by 30% through evaluation of learning efficiency and model behavior
- Experimented with Quantum Variational Circuits (QVC), reducing model parameter count by 80% while maintaining performance

Alet Systems April 2024 – December 2024
Software Engineer

- Tested and validated backend APIs to ensure system reliability and performance
- Debugged and resolved production level issues in backend systems, cutting response/resolution time by 40%
- Supported full-stack and mobile application maintenance

PROJECTS

Sift – AI Lead Qualification SaaS

- Built a multi-tenant AI agent that scores, enriches, and drafts replies to inbound leads in real time
- Developed the backend with NestJS and GraphQL, a FastAPI agent service, and a Next.js frontend, backed by PostgreSQL
- Implemented asynchronous job processing with Redis and BullMQ for real-time lead scoring and enrichment

Awaqi – Bilingual AI Tax Assistant

- Built a bilingual (Amharic/English) RAG assistant delivering cited, grounded answers to taxpayer questions from official tax proclamations Designed a hybrid retrieval pipeline (pgvector + keyword search, fused via RRF) with citations and confidence scoring to prevent hallucination
- Built "Awaqi Max," a ReAct agent (Gemini function calling) reasoning across the knowledge base and live web search for complex queries Automated document ingestion (PDF/HTML, OCR, chunking, embedding) and delivered the assistant via web chat, Telegram, and an admin dashboard

SKILLS

- **Programming:** Python, TypeScript, JavaScript,Go
- **Machine Learning & Data:** PyTorch, scikit-learn, pandas, NumPy
- **Backend & Data Engineering:** REST APIs, FastAPI, Node.js, Express.js, PostgreSQL, SQL, Airflow,GraphQl
- **Frontend:** Next.js
- **Tools & Technologies:** Git, Linux

CERTIFICATIONS & ACHIEVEMENTS

- A2SV Generative AI for Africa Hackathon Finalist (2023)
- Finalist – Reboot the Earth Hackathon (2024)
- Machine Learning Certification – AAiT (2023)
- Machine Learning Specialization (Deeplearning AI)